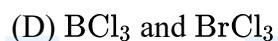
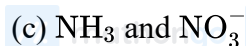
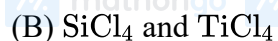
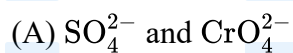


Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

Which of the following are isostructural pairs?



(1) A and C only

(2) A and B only

(3) B and C only

(4) C and D only

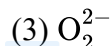
Q2: 24 Feb (Shift 2) - Single Correct

The correct shape and I-I-I bond angles respectively in I_3^- ion are :

(1) Trigonal planar; 120° (2) Distorted trigonal planar; 135° and 90° (3) Linear; 180° (4) T-shaped; 180° and 90°

Q3: 25 Feb (Shift 1) - Single Correct

According to molecular orbital theory, the species among the following that does not exist is:



Q4: 25 Feb (Shift 2) - Single Correct

Which among the following species has unequal bond lengths ?

Questions with Answer Keys

MathonGo



Q5: 26 Feb (Shift 1) - Single Correct

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation

Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical In the

light of the above statements, choose the most appropriate answer from the options given below :

(1) A is false but R is true

(2) Both A and R are true and R is the correct explanation of A

(3) A is true but R is false

(4) Both A and R are true and R is not the correct explanation of A

Q6: 26 Feb (Shift 2) - Single Correct

Match List-I with List-II.

List-I

(Molecule)

(a) Ne_2

(b) N_2

(c) F_2

(d) O_2

List-II

(Bond order)

(i) 1

(ii) 2

(iii) 0

(iv) 3

Choose the correct answer from the options given below:

(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (iii)

(2) (a) – (i), (b) – (ii), (c) – (iii), (d) – (iv)

(3) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

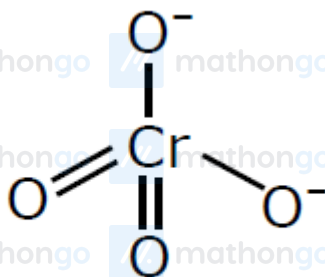
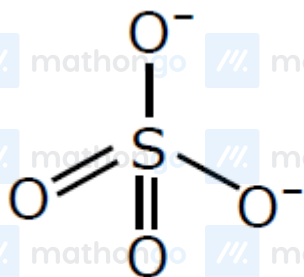
Questions with Answer Keys

MathonGo

(4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

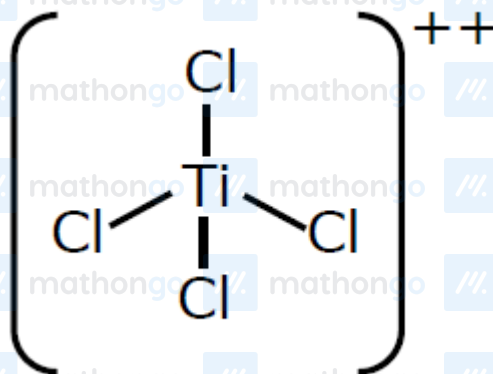
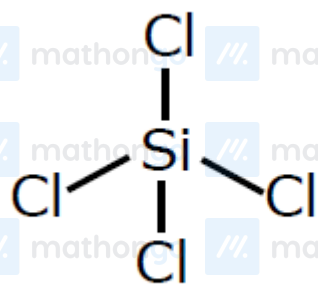
Answer Key**Q1 (2)****Q2 (3)****Q3 (4)****Q4 (4)****Q5 (3)****Q6 (1)**

Q1 (2)

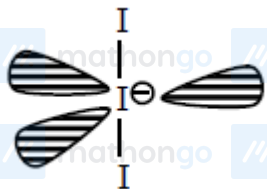
(a) SO_4^{2-} and CrO_4^{2-} both have tetrahedral structure.

Tetrahedral

Tetrahedral

(b) SiCl_4 and TiCl_4 both have Tetrahedral structure also.

Q2 (3)

 I_3^- sp^3 d hybridisation (2BP + 3L.P.) Linear geometry

Q3 (4)

B. O. of Be_2 is zero, So it does not exist.

Q4 (4)



sp^3d Hybridisation Sea-saw shape & axial bond length is more than equatorial bond length

Q5 (3)

Fact

Q6 (1)

Ne_2O $BO = 0$

N_2 $BO = 3$

F_2 $BO = 1$

O_2 $BO = 2$

As per molecular orbital theory